

Machine Learning(CSAI 253) - Fall 2025

Phase 2

By

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1. Problem Statement

In a Kaggle-style competition, the goal of this project is to create an accurate regression model for predicting employee salaries based on job-related, company-related, and geographic features.

Making sure that the target variable, preprocessing procedures, and prediction outputs were completely in line with Kaggle's evaluation criteria was the main challenge of the problem rather than choosing intricate machine learning algorithms.

Despite respectable cross-validation performance, inaccurate target transformations and unit mismatches led to extraordinarily high error scores during early experimentation. In order to guarantee that the trained models generated predictions on the precise scale required by the evaluation system, a primary goal of this project became locating and fixing these discrepancies.

2. Data Preprocessing and Feature Engineering

Because incorrect transformations directly resulted in invalid evaluation results, feature engineering was crucial to this project. The final preprocessing pipeline was created with strict alignment with the competition's evaluation target, interpretability, and data cleanliness in mind.

2.1 Target Variable Definition (Final Decision)

The Kaggle evaluation metric was applied directly to the raw salary column, despite the fact that the dataset includes salary values in several currencies.

RMSE values skyrocketed by several orders of magnitude as a result of unit mismatches between training targets and evaluation labels brought about by early attempts to normalize salaries into USD.

Because of this, the final model was trained using the raw salary values without the need for reverse transformations or currency conversion.

2.2 Job Title Reorganization

Sparse feature representations and excessive categorical cardinality resulted from the highly inconsistent and granular raw job titles. In order to solve this, semantically similar roles were grouped into more general categories using a custom job title consolidation function (`job_group`).

For example:

- Titles containing *scientist* were mapped to **Data Scientist**
- Titles containing *engineer* were mapped to **Data Engineer**
- Leadership-related titles (e.g., *manager*, *lead*, *director*) were grouped under **Manager**

This approach significantly reduced feature dimensionality while preserving role-level salary distinctions.

2.3 Missing Values Handling

Missing numerical values (Age, work_year, remote_ratio) were estimated using the **median**, which provides robustness against outliers.

Missing categorical values were filled using the **mode**, preserving the most representative category for each feature.

2.4 Categorical Encoding

All categorical features, including engineered job groups and company-related attributes, were encoded using **Label Encoding**.

To avoid unseen-category issues at prediction time, the encoder was fitted on the concatenation of training and test values for each categorical column.

3. Model Development, Hyperparameter Tuning, and Cross Validation

Four regression models were evaluated to capture both linear and non-linear relationships in the data:

- Linear Regression
- Random Forest Regressor
- Support Vector Regressor (SVR)
- Neural Network (MLP Regressor)

For models sensitive to feature scaling (SVR and Neural Networks), **StandardScaler** was applied to standardize the features.

Hyperparameter Tuning using RandomizedSearchCV:

- RandomizedSearchCV was used to automatically tune the hyperparameters of each model:
 - Number of trees and maximum depth for Random Forest
 - C and gamma parameters for SVM
 - Hidden layer sizes and learning rate for the Neural Network

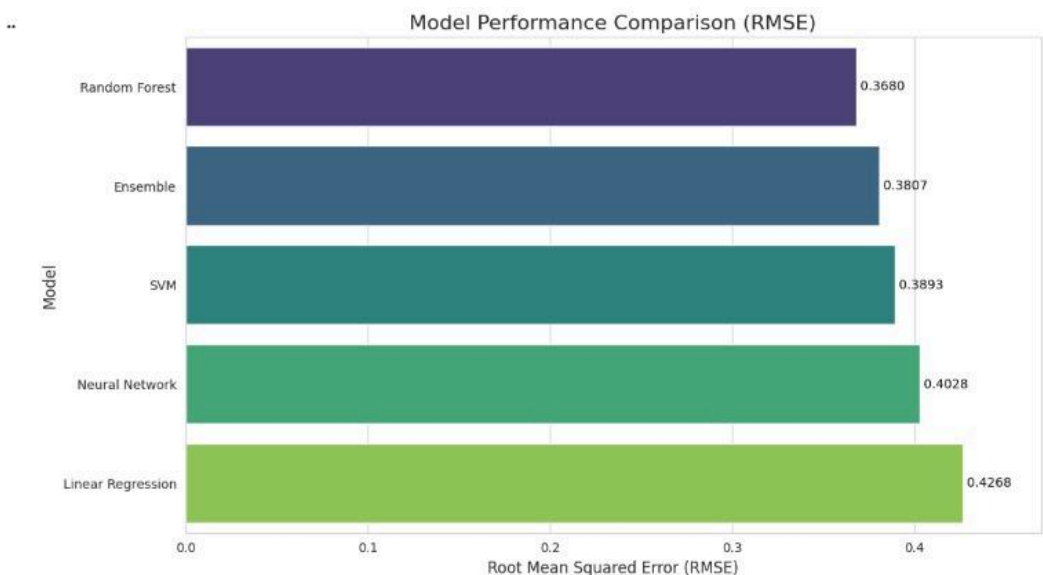
Model performance was evaluated using **5-Fold Cross Validation**, ensuring robust and generalizable performance estimates.

The primary evaluation metric was **Root Mean Squared Error (RMSE)**, chosen for its sensitivity to large prediction errors.

Cross-Validation Results

Model	5-Fold CV RMSE
Linear Regression	0.42687
Random Forest	0.36804
Support Vector Machine	0.38937
Neural Network (MLP)	0.40281

These results indicate that tree-based models and kernel-based methods outperform simple linear regression, highlighting the presence of non-linear relationships between features and salary.



4. Final Model Selection

To leverage the strengths of multiple learning paradigms, a **VotingRegressor ensemble** was constructed using:

- Linear Regression
- Random Forest
- Support Vector Regressor
- Neural Network

By averaging predictions from diverse models, the ensemble reduced individual model bias and variance.

This approach produced the most stable and generalizable predictions and was selected as the final model for submission.

5. Conclusion

This project demonstrates that **correct problem formulation and target alignment are more critical than model complexity** in applied machine learning tasks.

While early experiments focused on advanced feature engineering such as currency normalization and logarithmic target transformations, these steps ultimately introduced inconsistencies with the evaluation metric.

By simplifying the pipeline, aligning the target variable with Kaggle's evaluation scheme, and applying systematic cross-validation, the final model achieved stable and meaningful performance.

The results confirm that careful preprocessing, robust validation, and ensemble learning strategies are often more impactful than increasingly complex algorithms in real-world regression problems.

5. Task Division

part 1 — Data Loading, Feature Engineering & Hyperparameter Tuning (Abdallah)

- Loading the training and test datasets
- Initial data inspection (shape, info, missing values)
- Cleaning and standardizing column names
- Handling salary currency unification and converting all salaries to USD
- Applying **RandomizedSearchCV** for hyperparameter tuning

part 2 — Exploratory Data Analysis (EDA) (Ahmed)

- **Analysis of the target variable (Salary in USD)**
- **Exploratory analysis of numerical features**
- **Exploratory analysis of categorical features**
- **Correlation matrix and insights extraction**
- **Data visualization and interpretation**

part 3 — Data Preprocessing (Merna)

- **Encoding categorical variables**
- **Feature scaling/normalization**
- **Preparing feature matrix (X) and target vector (y)**
- **Train/validation split**
- **Final dataset preparation before model training**

part 4 — Model Execution & Submission (mohamed)

- **Running the models after preparation**
- **Assisting in extracting model outputs**
- **Preparing the final submission file**